

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)**

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-07	Team No-13	Project Title-Job Portal with Resume Parsing & Matching
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1. Abstract

The **Job Portal with Resume Parsing & Matching** is a database-driven web application designed to simplify and improve the recruitment process for job seekers and recruiters. The system provides a platform where candidates can create profiles, upload resumes, search for job opportunities, and apply for suitable positions. Recruiters can create company profiles, post job vacancies, view applications, and identify suitable candidates efficiently.

A key feature of the system is **resume parsing**, which automatically extracts important information such as skills, educational qualifications, work experience, projects, and certifications from uploaded resumes. The extracted information is stored in a structured relational database for easy retrieval and analysis. The system then compares candidate profiles with job requirements using a matching algorithm and generates a compatibility score based on factors such as skills, experience, education, and certifications.

The project uses **Database Systems and Engineering** concepts such as relational database design, normalization, primary and foreign keys, constraints, indexing, SQL queries, transactions, and query optimization. For **distributed backend development**, the system is divided into services such as user management, job management, resume processing, matching, and application management. These services communicate through REST APIs, improving scalability, maintainability, and performance.

The proposed system reduces manual resume screening, provides personalized job recommendations, helps recruiters identify suitable candidates quickly, and improves the overall efficiency of the recruitment process. Thus, the project demonstrates the practical application of database management and distributed backend technologies in a real-world recruitment system.

2. Team Details

Roll Number(s)	Name of the Student	Signature
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3. Frameworks / Platforms / Software Used

The project will use **PostgreSQL with pgvector** as the primary relational database for storing users, jobs, resumes, skills, applications, and matching data, covering relational database design, SQL, normalization, indexing, transactions, and vector similarity search. **MongoDB** will be used for selected document-based data to demonstrate SQL and NoSQL integration. **Python with FastAPI** will be used to develop RESTful APIs, with **Pydantic, SQLAlchemy, JWT, and OAuth2** for validation, database integration, and authentication. **Node.js with Express** and **Spring Boot** can be used for selected microservices to demonstrate multi-framework backend engineering. **Redis** will support caching and distributed services, while **Docker and Docker Compose** will be used for containerization and multi-service deployment. **Postman and Pytest** will be used for API and backend testing. **Git and GitHub** will be used for version control and CI/CD using **GitHub Actions**. **OpenTelemetry, Prometheus, and Grafana** will be used for observability and monitoring, with **Kubernetes** considered for basic deployment awareness. **Visual Studio Code** and **pgAdmin** will be used as development and database management tools.

Guide Name and Signature (remarks if any) _____